VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum, Karnataka -590014

LAB REPORT
on

ANALYSIS AND DESIGN OF ALGORITHMS

(23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
Computer Science and Engineering

B.M.S. COLLEGE OF ENGINEERING
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CERTIFICATE

This is to certify that the Lab work entitled ***"ANALYSIS AND DESIGN OF ALGORITHMS",*** carried out by Pooja Yoganarasimha (1BF24CS217), who is bonafide student of ***B.M. S. College of Engineering***. It is in partial fulfillment for the award of ***Bachelor of Engineering in Computer Science and Engineering*** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of ***Analysis and Design of Algorithms Lab - (23CS4PCADA)*** work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1: Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>
#define MAX 10
int main() {
   intn,i,j;
   int indegree[MAX] = {0};
   int graph[MAX][MAX];int
   order[MAX], count = 0;
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter adjacency matrix:\n");
   for(i = 0; i < n; i++) {
      for(j = 0; j < n; j++) {
         scanf("%d", &graph[i][j]);
         if(graph[i][j] == 1) {
            indegree[j]++;
         }
      }
   }
   while(count < n) {
      int found = 0;
      for(i = 0; i < n; i++) {
         if(indegree[i] == 0){
            order[count++] = i;
            indegree[i] = -1;
            for(j = 0; j < n; j++) {
               if(graph[i][j]==1){
                  indegree[j]--;
```

```
                }
            }
            found=1;
        }
    }
    if(!found)break;
}
if(count<n){
    printf("NoTopologicalSequence(cycledetected)\n");
}else{
    printf("Topological Order: ");
    for(i = 0; i < n; i++) {
        printf("%d",order[i]);
    }
    printf("\n");
}
return0;
}
```

Output:

Labprogram2:
Implement Johnson Trotter algorithm to generate permutations.

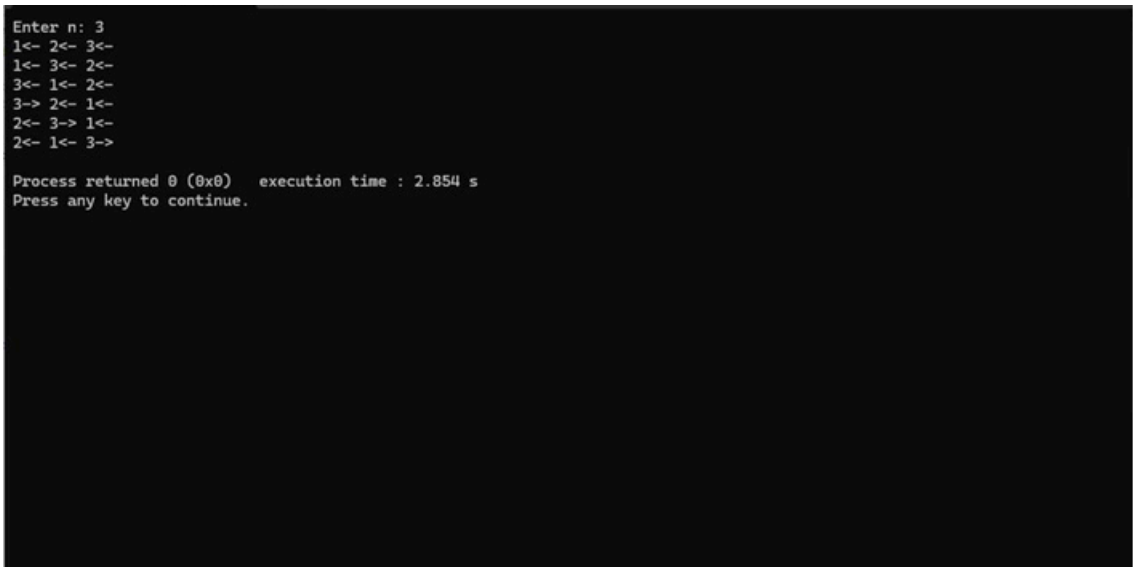
```
#include <stdio.h>
#define LEFT 0
#defineRIGHT1
void print(int a[], int dir[], int n) {
   for(int i = 0; i < n; i++) {
      if(dir[a[i]] == LEFT)
         printf("%d<- ", a[i]);
      else
         printf("%d->",a[i]);
   }
   printf("\n");
}
int getMobile(int a[], int dir[], int n) {
   int mobile = 0;
```

```
    for(inti=0;i<n;i++){

        if(dir[a[i]] == LEFT && i > 0) {
            if(a[i] > a[i-1] && a[i] > mobile)
                mobile=a[i];
        }

        if(dir[a[i]] == RIGHT && i < n-1) {
            if(a[i] > a[i+1] && a[i] > mobile)
                mobile=a[i];
        }
    }
    returnmobile;
}

int findPos(int a[], int n, int mobile) {
    for(int i = 0; i < n; i++) {
        if(a[i] == mobile)
            return i;
    }
    return-1;
}

void johnsonTrotter(int n) {
    int a[n], dir[n+1];


    for(int i = 0; i < n; i++) {
        a[i] = i + 1;
```

```
    dir[i+1]=LEFT;
}

print(a,dir,n);

while(1){
    intmobile=getMobile(a,dir,n);

    if(mobile== 0)
        break;

    intpos=findPos(a,n,mobile);


    if(dir[mobile] == LEFT) {
        int temp = a[pos];
        a[pos] = a[pos-1];
        a[pos-1]=temp;
        pos = pos - 1;
    }else{
        int temp = a[pos];
        a[pos] = a[pos+1];
        a[pos+1] = temp;
        pos = pos + 1;
    }

    for(int i = 1; i <= n; i++) {
        if(i > mobile) {
            dir[i]=!dir[i];
```

```
            }
        }

        print(a,dir,n);
    }
}

intmain(){
    int n;
    printf("Entern:");
    scanf("%d",&n);

    johnsonTrotter(n);

    return0;
}
```

Output:



Lab program 3: Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include    <stdio.h>
#include   <stdlib.h>
#include <time.h>


//Mergefunction
voidmerge(intarr[],intlb,intmid,intub){


   int n1 = mid - lb + 1;
   int n2 = ub - mid;


   intL[n1],R[n2];


   //Copydata
   for (inti = 0; i < n1; i++)
       L[i] = arr[lb + i];


   for (int j = 0; j < n2; j++)
        R[j]=arr[mid+1+j];


   inti=0,j=0,k=lb;


   //Mergearrays
   while(i<n1&&j<n2){


      if(L[i]<=R[j])
          arr[k++]=L[i++];
```

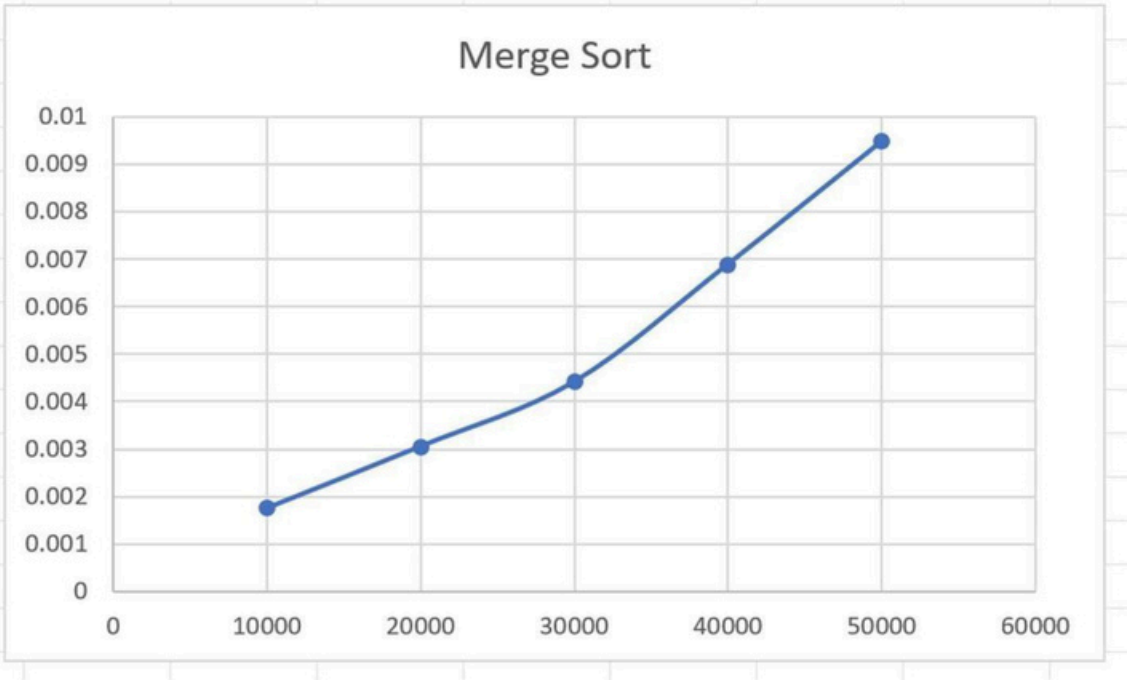
```
        else
            arr[k++]=R[j++];
    }

    // Remaining elements
    while (i < n1)
        arr[k++]=L[i++];

    while (j < n2)
        arr[k++] = R[j++];
}

//MergeSort
voidmergeSort(intarr[],intlb,intub){

    if(lb<ub){

        intmid=(lb+ub)/2;

        mergeSort(arr, lb, mid);
        mergeSort(arr, mid + 1, ub);

        merge(arr,lb,mid,ub);
    }
}

intmain(){

    intn;
```

```
printf("Enter number of elements: ");
scanf("%d", &n);

intarr[n];

printf("Enterelements:\n");

 for (inti = 0; i < n; i++) {
     scanf("%d",&arr[i]);
}

clock_tstart=clock();

mergeSort(arr,0,n-1);

clock_tend=clock();

doubletime_taken=
    (double)(end-start)/CLOCKS_PER_SEC;

printf("\nSortedarray:\n");

for (inti = 0; i < n; i++) {
    printf("%d ", arr[i]);
}

printf("\n");
```

```
    printf("Timetaken:%.6fseconds\n",time_taken);

    return0;
}
```

Output:

```
Enter number of elements: 30000
Merge Sort completed.
Time taken: 0.002000 seconds

Process returned 0 (0x0)   execution time : 2.444 s
Press any key to continue.
```



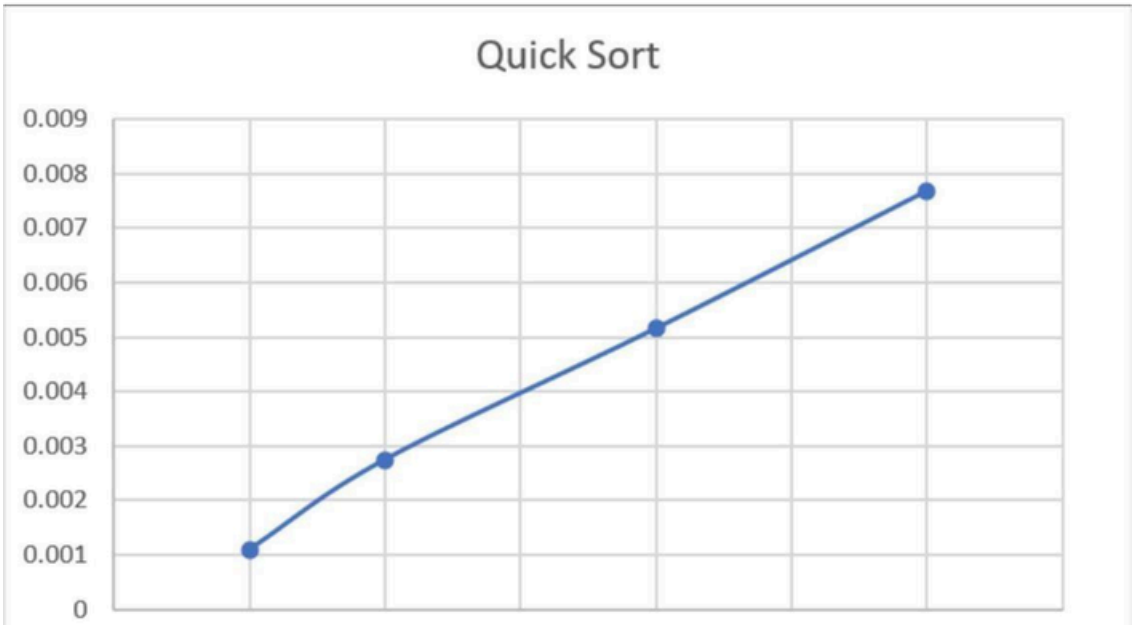
Lab program 4: Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include <stdio.h>
#include<stdlib.h>
#include <time.h>

//Functiontoswaptwoelements
void swap(int *a, int *b) {
    inttemp=*a;
    *a=*b;
    *b=temp;
}

//Partitionfunction
intpartition(intarr[],intlow,inthigh){
    intpivot=arr[high];//choosinglastelementaspivot int i =
    (low - 1);

    for(intj=low;j<high;j++){ if
        (arr[j] <= pivot) {
            i++;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[i+1],&arr[high]); return
    (i + 1);
}

//QuickSortfunction
voidquickSort(intarr[],intlow,inthigh){ if
    (low < high) {
        intpi=partition(arr,low,high);
```

```
        quickSort(arr, low, pi - 1);
        quickSort(arr,pi+1,high);
    }
}

intmain(){
    int N;
    printf("Enternumberofelements:");
    scanf("%d", &N);

    intarr[N];
    printf("Enter the elements:\n");
    for (int i = 0; i < N; i++) {
        scanf("%d",&arr[i]);
    }

    clock_tstart,end;
    start = clock();

    quickSort(arr,0,N-1);

    end=clock();

      printf("Sorted array:\n");
      for(int i = 0; i < N; i++) {
          printf("%d ", arr[i]);
    }

    doubletime_taken=((double)(end- start)) /CLOCKS_PER_SEC;
    printf("\nTime taken: %f seconds\n", time_taken);

    return0;
}
```

Output:

```
Enter number of elements: 10000
Quick Sort completed.
Time taken: 0.001000 seconds

Process returned 0 (0x0)   execution time : 3.180 s
Press any key to continue.
```



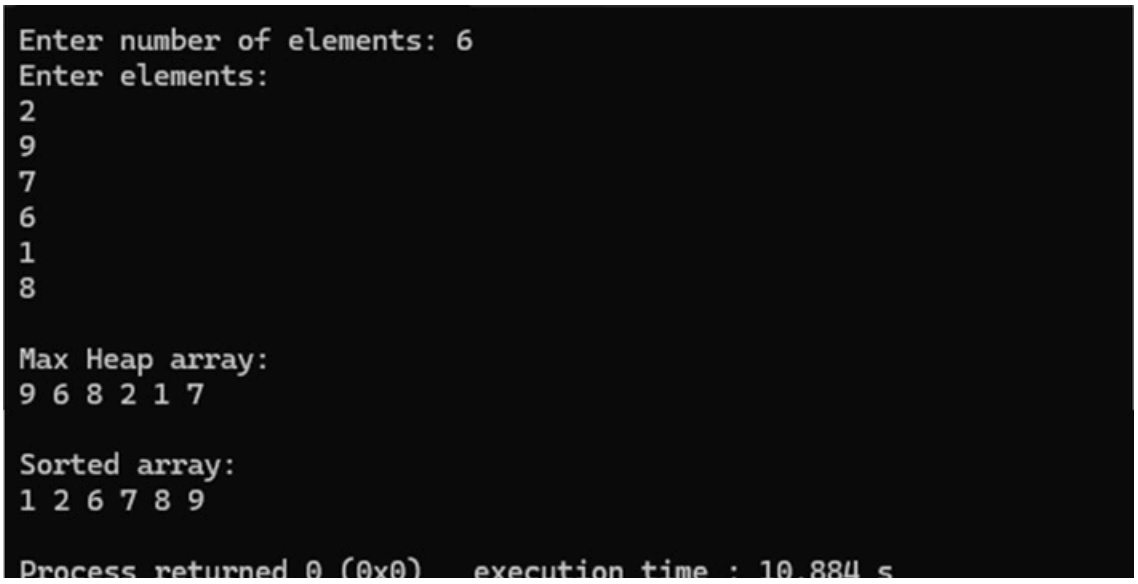
Lab program 5: Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include<stdio.h>
void swap(int *a, int *b) {
    int temp = *a;
    *a=*b;
 *b=temp;
}
void siftUp(int a[], int i) {
    while(i > 0) {
        int parent= (i -1) / 2;
        if(a[parent] < a[i]) {
            swap(&a[parent], &a[i]);
            i = parent;
        }else{
            break;
        }
    }
}

// Build heap using Top-Down approach
void buildHeap(int a[], int n) {
    for(int i = 1; i < n; i++) {
        siftUp(a, i);
    }
}
void heapify(int a[], int n, int i) {
    int largest = i;
```

```
    int left = 2*i + 1;
    int right= 2*i + 2;

    if(left < n && a[left] > a[largest])
        largest = left;
    if(right < n && a[right] > a[largest])
        largest = right;
    if(largest!=i){
        swap(&a[i], &a[largest]);
        heapify(a, n, largest);
    }
}
 void heapSort(int a[], int n) {
    buildHeap(a, n);
    for(inti=n-1;i>0;i--){
        swap(&a[0], &a[i]);
        heapify(a, i, 0);
    }
}
 void print(int a[], int n) {
    for(inti=0;i<n;i++)
        printf("%d ", a[i]);
    printf("\n");
}
 intmain(){
    int n;
    printf("Entern:");
    scanf("%d",&n);
```

```
    inta[n];
    printf("Enter elements:\n");
    for(int i = 0; i < n; i++)
    scanf("%d", &a[i]);
    heapSort(a, n);
    printf("Sorted array:\n");
    print(a, n);
    return0;
}
```

Output:



Lab program6:
Implement 0/1 Knapsack problem using dynamic programming.

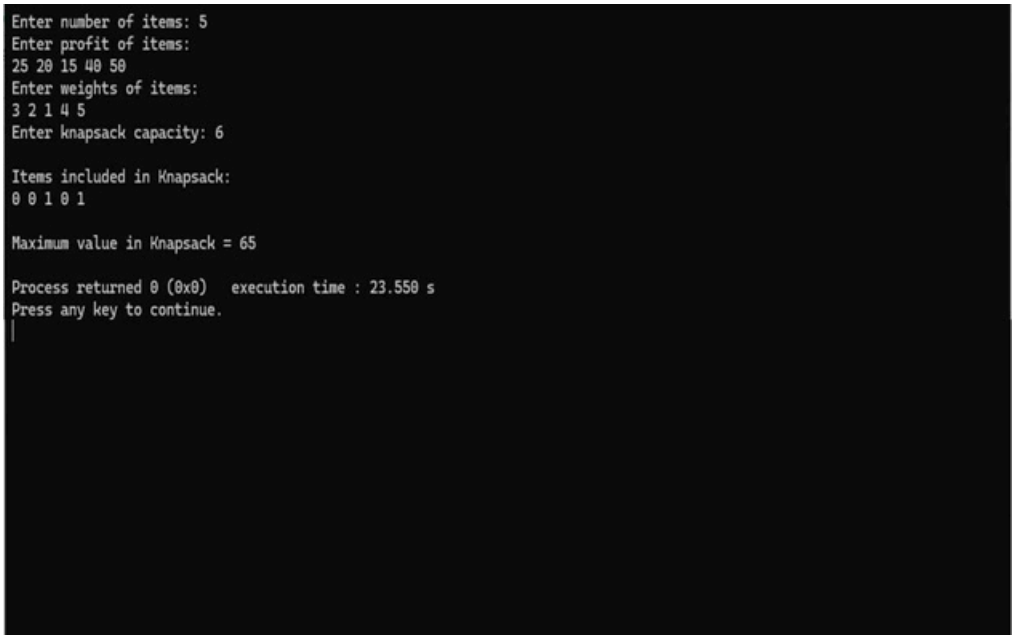
```
#include<stdio.h>

int knapsack(int W, int wt[], int val[], int n) {
   int dp[n+1][W+1];

   //BuildDPtable
   for(inti=0;i<=n;i++){
      for(intw=0;w<=W;w++){

         if (i == 0 || w== 0){
            dp[i][w] = 0;
         }

         elseif(wt[i-1]<=w){

            int include = val[i-1] + dp[i-1][w - wt[i-1]];
            int exclude = dp[i-1][w];

            dp[i][w]=(include>exclude)?include:exclude;
         }

         else{
            dp[i][w]=dp[i-1][w];
         }
      }
   }
```

```
// Print selected items
int selected[n];

for (inti = 0; i < n; i++)
    selected[i] = 0;

int i = n;
intw=W;

while(i>0&&w>0){

    //Itemincluded
    if (dp[i][w] != dp[i-1][w]) {
        selected[i-1] = 1;
        w=w-wt[i-1];
    }

    i--;
}

printf("\nItemsincludedinKnapsack:\n");

for (int i = 0; i < n; i++) {
    printf("%d ", selected[i]);
}

printf("\n");

returndp[n][W];
```

```
}

intmain(){

   intn,W;

   printf("Enter number of items: ");
   scanf("%d", &n);

   intval[n],wt[n];

   printf("Enter profit of items:\n");
   for (int i = 0; i < n; i++) {
      scanf("%d",&val[i]);
   }

   printf("Enter weights of items:\n");
   for (int i = 0; i < n; i++) {
      scanf("%d",&wt[i]);
   }

   printf("Enter knapsack capacity: ");
   scanf("%d", &W);

   intresult=knapsack(W,wt,val,n);

   printf("\nMaximumvalueinKnapsack=%d\n",result);

   return0;
```

}

Output:


Enter number of items: 5
Enter profit of items:
25 20 15 40 50
Enter weights of items:
3 2 1 4 5
Enter knapsack capacity: 6
Items included in Knapsack:
0 0 1 0 1
Maximum value in Knapsack = 65
Process returned 0 (0x0) execution time : 23.550 s
Press any key to continue.

Lab program7:
Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include <stdio.h>
#defineINF99999
#defineMAX100
void printSolution(int dist[MAX][MAX], int V) {
   printf("\nShortest distance matrix:\n");
    for (int i = 0; i < V; i++) {for
        (int j = 0; j < V; j++) {
          if(dist[i][j]==INF)
             printf("%7s", "INF");
          else
             printf("%7d",dist[i][j]);
       }
       printf("\n");
   }
}
void floydWarshall(int graph[MAX][MAX], int V) {
   int dist[MAX][MAX];
   for (int i = 0; i < V; i++)for
   (int j = 0; j < V; j++)
         dist[i][j]=graph[i][j];

   //Floyd-Warshall
   for (int k = 0; k < V; k++) {
      for(inti=0;i<V;i++){
         for(intj=0;j<V;j++){
            if (dist[i][k] < INF && dist[k][j] < INF &&
               dist[i][k] + dist[k][j] < dist[i][j]) {
```

```
                dist[i][j]=dist[i][k]+dist[k][j];
            }
          }
        }   }
    for (int i = 0; i < V; i++) {
        if (dist[i][i] < 0) {
            printf("\nNegative cycle detected!\n");
            return;
        }
    }
    printSolution(dist,V);
}
intmain(){
    int V;
    int graph[MAX][MAX];
    printf("Enter number of vertices: ");
    scanf("%d", &V);
    printf("\nEnter adjacency matrix (-1 for no edge):\n");
    for (int i = 0; i < V; i++) {
        for (int j = 0; j < V; j++) {
            scanf("%d", &graph[i][j]);
            if (graph[i][j] == -1)
                graph[i][j]=INF;
        }
    }
    for (inti = 0; i < V; i++)
        graph[i][i] = 0;
    floydWarshall(graph, V);
    return 0;}
```

Output:

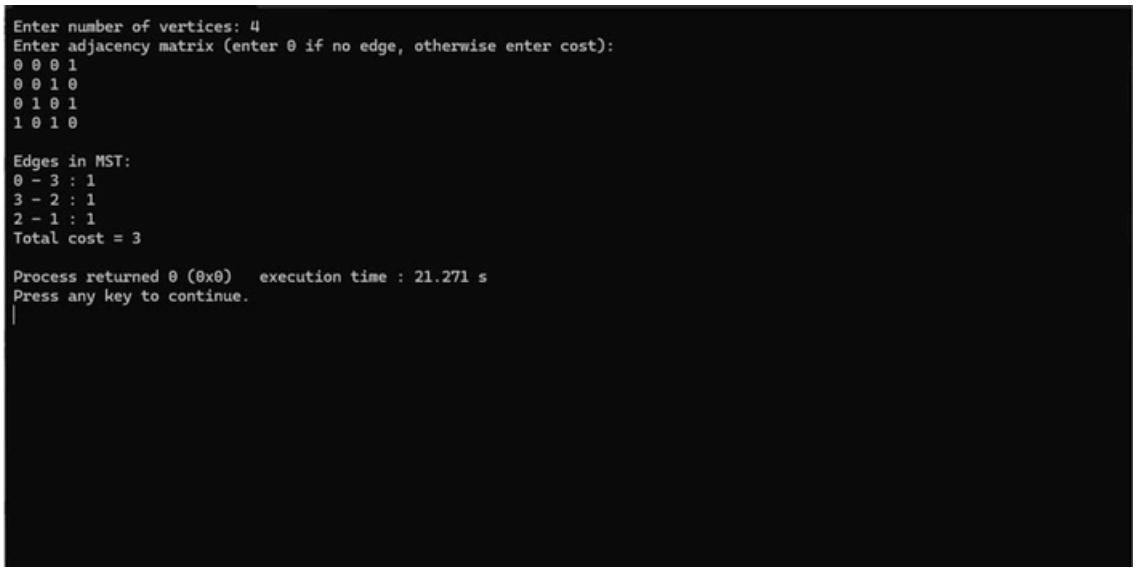

Enter number of vertices: 4
Enter adjacency matrix (-1 for no edge):
0 3 -1 5
2 0 -1 4
-1 6 0 -1
-1 -1 2 0
Shortest distance matrix:
0 3 7 5
2 0 6 4
8 6 0 10
10 8 2 0
Process returned 0 (0x0) execution time : 30.376 s
Press any key to continue.

Lab program 8a: Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>
#include <limits.h>
int minKey(int key[], int mstSet[], int V) {
   int min = INT_MAX, min_index = -1;
   for (int v = 0; v < V; v++)
      if (mstSet[v] == 0 && key[v] < min)
         min = key[v], min_index = v;
   returnmin_index;
}
void primMST(int graph[50][50], int V) {
   int parent[V], key[V], mstSet[V];
   inttotalCost=0;

   for (int i = 0; i < V; i++) {
      key[i] = INT_MAX;
      mstSet[i] = 0;
   }
   key[0]=0;
   parent[0]=-1;
   for (int count = 0; count < V - 1; count++) {
   int u = minKey(key, mstSet, V);mstSet[u]
   = 1;

      for(intv=0;v<V;v++)
         if (graph[u][v] && mstSet[v] == 0 && graph[u][v] < key[v]) {
            parent[v] = u;
            key[v]=graph[u][v];
```

```
            }
        }

        printf("\nEdges in MST:\n");
        for (int i = 1; i < V; i++) {
            printf("%d - %d : %d\n", parent[i], i, graph[i][parent[i]]);
            totalCost += graph[i][parent[i]];
        }
        printf("Totalcost=%d\n",totalCost);
}
intmain(){
    int V;
    intgraph[50][50];
    printf("Enter number of vertices: ");
    scanf("%d", &V);
    printf("Enter adjacency matrix (enter 0 if no edge, otherwise enter cost):\n");
    for (int i = 0; i < V; i++)
        for (int j = 0; j < V; j++)
            scanf("%d", &graph[i][j]);

    primMST(graph,V);

    return0;
}
```

Output:


Enter number of vertices: 4
Enter adjacency matrix (enter 0 if no edge, otherwise enter cost):
0 0 0 1
0 0 1 0
0 1 0 1
1 0 1 0
Edges in MST:
0 - 3 : 1
3 - 2 : 1
2 - 1 : 1
Total cost = 3
Process returned 0 (0x0) execution time : 21.271 s
Press any key to continue.

Lab program 8b: Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>
#include <stdlib.h>
struct Edge {
intsrc,dest,weight;
};
structSubset{ int
parent;
};
int find(struct Subset subsets[],int i){
while(subsets[i].parent!=i)
i=subsets[i].parent;
returni;
}
void Union(struct Subset subsets[],int x,int y){
int xroot=find(subsets,x);
int yroot=find(subsets,y);
subsets[xroot].parent=yroot;
}
intcompare(constvoid*a,constvoid*b){
return((structEdge*)a)->weight-((structEdge*)b)->weight;
}
void KruskalMST(struct Edge edges[],int V,int E){
struct Edge result[V];
struct Subset subsets[V];
qsort(edges,E,sizeof(struct Edge),compare);
for(int i=0;i<V;i++)
subsets[i].parent=i;
```

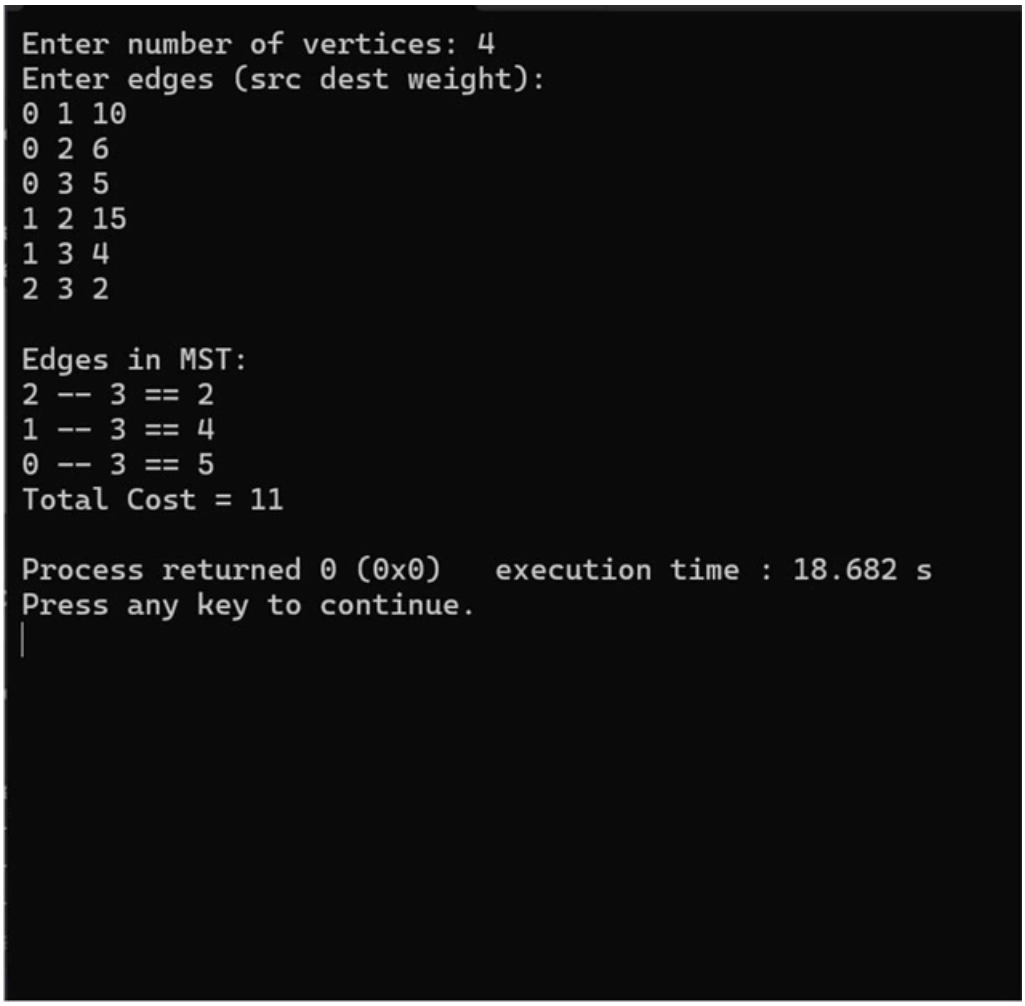
```
inte=0;
inti=0;
while(e<V-1&&i<E){
struct Edge next=edges[i++];
int x=find(subsets,next.src);
int y=find(subsets,next.dest);
if(x!=y){
result[e++]=next;
Union(subsets,x,y);
}
}
printf("\nEdges in MST:\n");
int total=0;
for(i=0;i<e;i++){
printf("%d -- %d == %d\n",result[i].src,result[i].dest,result[i].weight);
total+=result[i].weight;
}
printf("TotalCost=%d\n",total);
}
intmain(){
int V;
printf("Enter number of vertices: ");
scanf("%d",&V);
int E=V*(V-1)/2;
struct Edge edges[E];
printf("Enter edges (src dest weight):\n");
for(int i=0;i<E;i++){
scanf("%d%d%d",&edges[i].src,&edges[i].dest,&edges[i].weight);
}
```

KruskalMST(edges,V,E);

return 0;

}

Output:

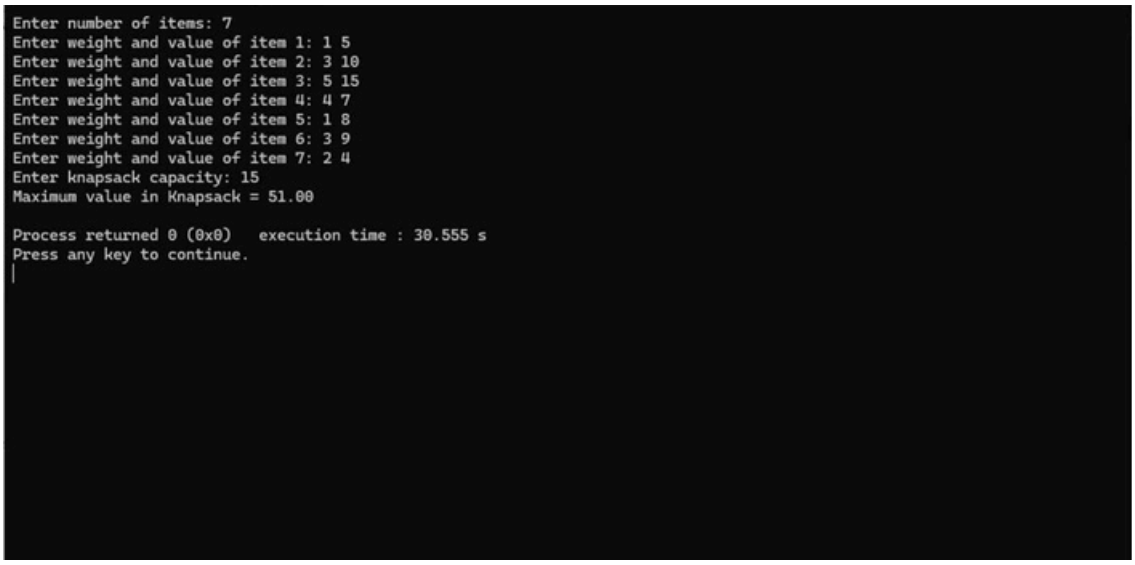


Lab program9: Implement Fractional Knapsack using Greedy technique

```
#include<stdio.h>
struct Item{int weight;int value;};
int main(){
   intn,capacity;
   printf("Enter number of items: ");
   scanf("%d",&n);
   struct Item items[n];
   for(int i=0;i<n;i++){
      printf("Enter weight and value of item %d: ",i+1);
      scanf("%d%d",&items[i].weight,&items[i].value);
   }
   printf("Enter knapsack capacity: ");
   scanf("%d",&capacity);
   doubleratio[n];
   for(int i=0;i<n;i++)ratio[i]=(double)items[i].value/items[i].weight;
   for(int i=0;i<n-1;i++){
      for(int j=i+1;j<n;j++){
         if(ratio[i]<ratio[j]){
            double tempR=ratio[i];ratio[i]=ratio[j];ratio[j]=tempR;
            struct Item temp=items[i];items[i]=items[j];items[j]=temp;
         }
      }
   }
   double maxValue=0.0;int remaining=capacity;
   for(int i=0;i<n;i++){
      if(items[i].weight<=remaining){
         remaining-=items[i].weight;
```

```
            maxValue+=items[i].value;
        }else{
            maxValue+=ratio[i]*remaining;
            break;
        }
    }
    printf("Maximum value in Knapsack = %.2f\n",maxValue);
    return 0;
}
```

Output:



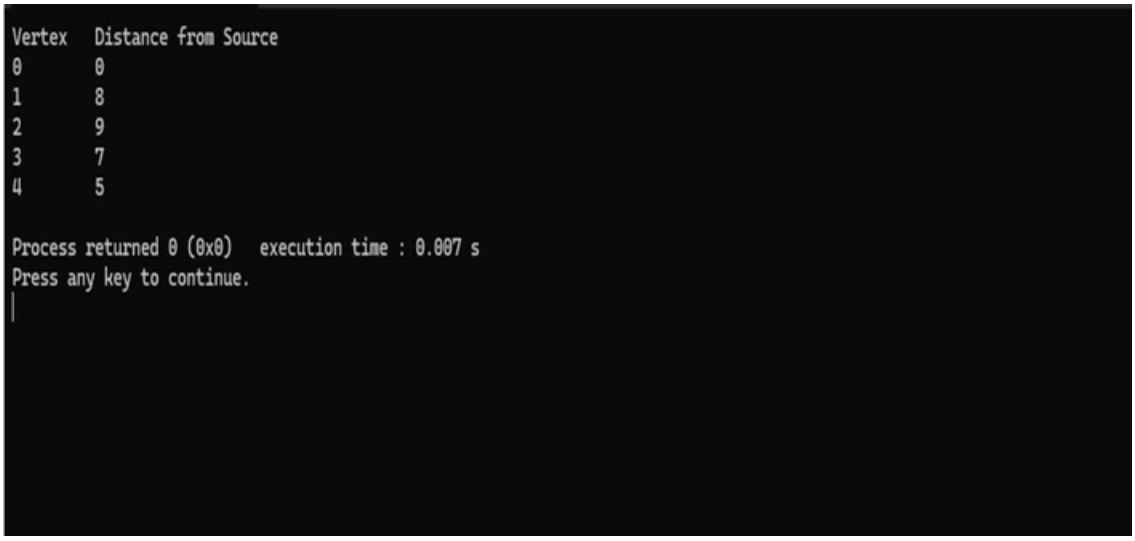
Lab program 10: From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include <stdio.h>
#include <limits.h>
#defineV10//maximumnumberofvertices
int minDistance(int dist[], int sptSet[], int vertices) {
   int min = INT_MAX, min_index;
   for(intv=0;v<vertices;v++){
      if (sptSet[v] == 0 && dist[v] <= min) {
         min  =  dist[v];
         min_index=v;
      }
   }
   returnmin_index;
}
void printSolution(int dist[], int vertices, int src) {
   printf("Vertex \t Distance from Source %d\n", src);
   for (int i = 0; i < vertices; i++) {
      printf("%d\t%d\n",i,dist[i]);
   }
}

void dijkstra(int graph[V][V], int src, int vertices) {
   int dist[vertices];       // shortest distance from src
   intsptSet[vertices];//whethervertexisincludedinshortestpathtree

   for (int i = 0; i < vertices; i++) {
      dist[i] = INT_MAX;sptSet[i];
```

```
    }
    dist[src]=0;
    for (int count = 0; count < vertices - 1; count++) {
        int u = minDistance(dist, sptSet, vertices);
        sptSet[u] = 1;
        for(intv=0;v<vertices;v++){
            if (!sptSet[v] && graph[u][v] && dist[u] != INT_MAX
                && dist[u] + graph[u][v] < dist[v]) {
                dist[v]=dist[u]+graph[u][v];
            }
        }
    }
    printSolution(dist,vertices,src);
}
int main()
    {intvertices
    ;
    printf("Enter number of vertices: ");
    scanf("%d", &vertices)
    intgraph[V][V];
    printf("Enter adjacency matrix (0 if no edge):\n");
    for (int i = 0; i < vertices; i++) {
        for (int j = 0; j < vertices; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    intsrc;
    printf("Enter source vertex: ");
    scanf("%d", &src);
```

```
    dijkstra(graph,src,vertices);

    return0;
}
```

Output:



Lab program 11: *Implement "N-Queens Problem" using Backtracking*.

```
#include <stdio.h>
#include <stdbool.h>
#define MAX 20
intN;
int board[MAX][MAX];
bool isSafe(int row,int col){
   for(int i=0;i<row;i++){
      if(board[i][col])return false;
      if(col-(row-i)>=0&&board[i][col-(row-i)])return false;
      if(col+(row-i)<N&&board[i][col+(row-i)])returnfalse;
   }
   returntrue;
}
void printBoard(){
   for(int i=0;i<N;i++){
      for(int j=0;j<N;j++)printf("%c ",board[i][j]?'Q':'.');
      printf("\n");
   }
   printf("\n");
}
void solve(int row){
   if(row==N){
      printBoard();
      return;
   }
   for(int col=0;col<N;col++){
      if(isSafe(row,col)){
         board[row][col]=1;
```

```
            solve(row+1);
            board[row][col]=0;
        }
    }
}
int main(){
    printf("Enter the size of the board (N): ");
    scanf("%d",&N);
    if(N>MAX||N<1){
        printf("Invalid board size. Please enter between 1 and %d.\n",MAX);
        return 1;
    }
    solve(0);
    return0;
}
```

Output:

LEET CODE PROBLEMS

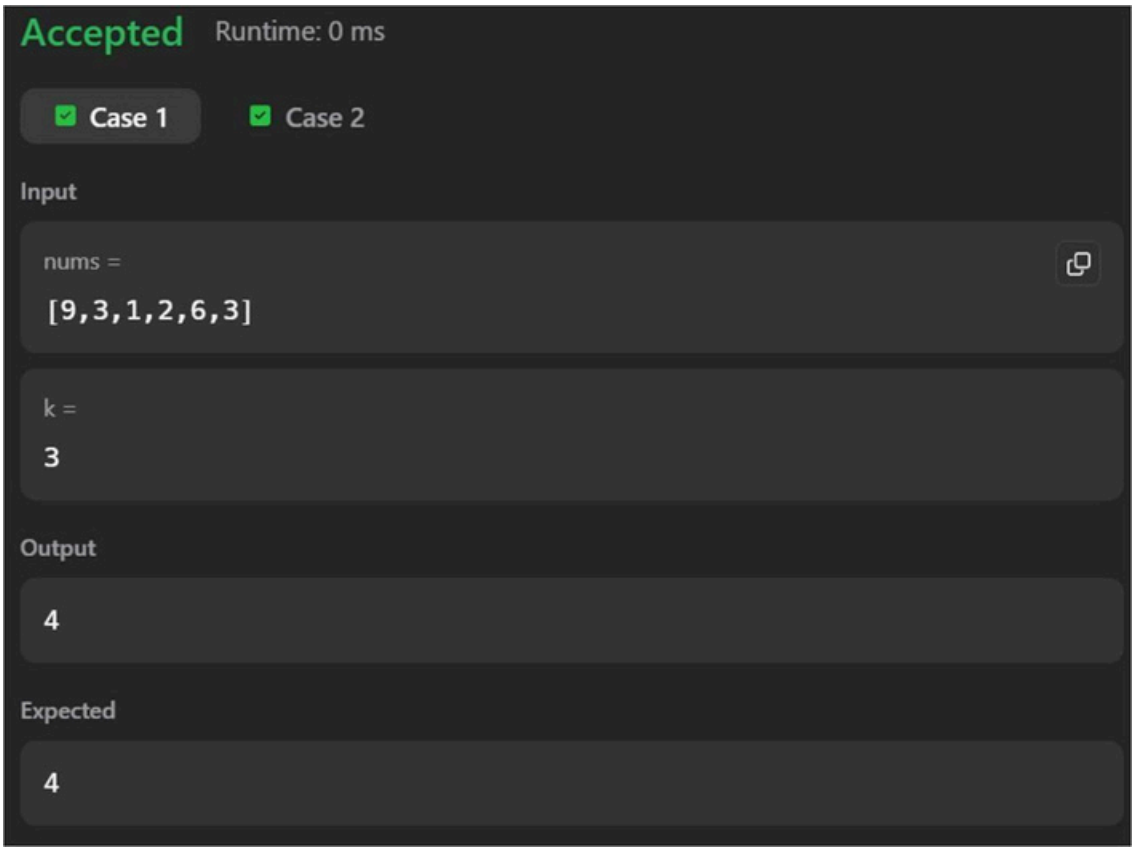
Leetcode 3658: GCD of odd and even sums

```
intgcdOfOddEvenSums(intn){

    intoddSum=n*n;
    intevenSum=n*(n+1);

    // find gcd using Euclidean algorithm
    while (evenSum != 0) {
        inttemp=evenSum;
        evenSum = oddSum % evenSum;
        oddSum = temp;
    }

    returnoddSum;
}
```

Leetcode 2447 : No of subarrays with GCD equal to k

```
 int gcd(int a, int b) {
    while (b != 0) {
       inttemp=b;   b
       =   a   %   b;
       a=temp;
    }
    returna;
}

        intsubarrayGCD(int*nums,intnumsSize,intk){

    intcount=0;

    for(inti=0;i<numsSize;i++){

       intcurrentGCD=0;

       for(intj=i;j<numsSize;j++){

             currentGCD=gcd(currentGCD,nums[j]);

          if (currentGCD == k) {
             count++;
          }
          else if (currentGCD < k) {
             break;
          }
```

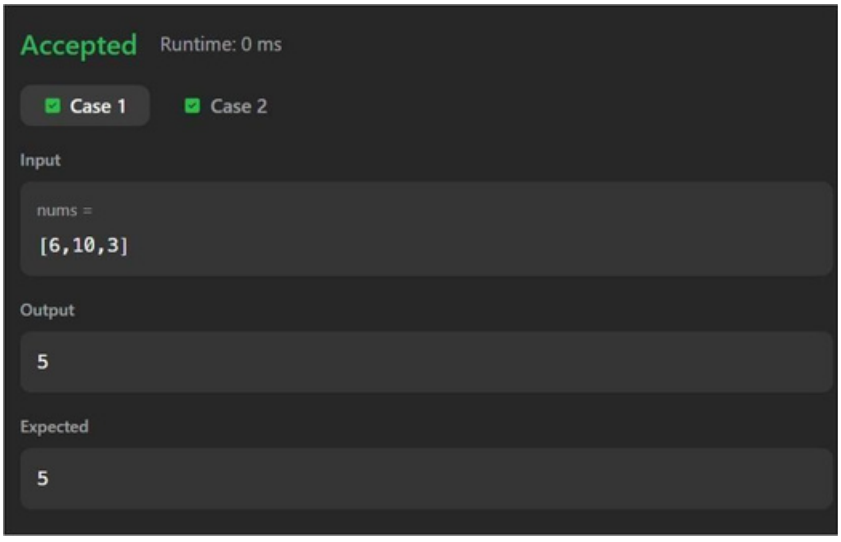
```
        }
    }

    returncount;
}
```



Leetcode 1819: No of different subsequences GCDs

```
int gcd(int a, int b) {
while (b != 0) {
        inttemp=b;    b
        =    a    %    b;
        a=temp;
    }
    returna;
}

intcountDifferentSubsequenceGCDs(int*nums,intnumsSize){

    intmaxNum=0;

    for (int i = 0; i < numsSize; i++) {
        if (nums[i] > maxNum) {
            maxNum=nums[i];
        }
    }

    int*exists=(int*)calloc(maxNum+1,sizeof(int));

    for (int i = 0; i < numsSize; i++) {
        exists[nums[i]] = 1;
    }

    intanswer=0;

    for(intx=1;x<=maxNum;x++){
```

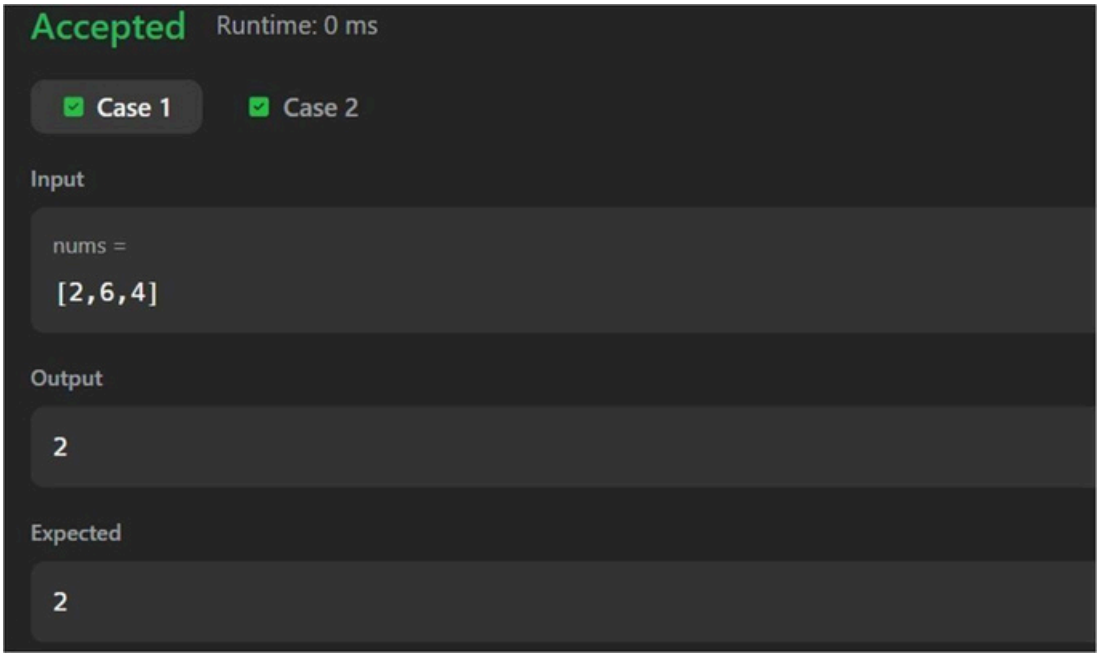
```
    intcurrentGCD=0;
   for(intmultiple=x;multiple<=maxNum;multiple+=x){

      if(exists[multiple]){
         currentGCD=gcd(currentGCD,multiple);

         if (currentGCD == x) {
            answer++;
            break;
         }
      }
   }
}

free(exists);
returnanswer;
}
```



Leetcode 3867 : Sum of GCD of formed pairs

```
 int gcd(int a, int b) {
    while (b != 0) {
       inttemp=b;   b
       =   a   %   b;
       a=temp;
    }
    returna;
}
int cmp(const void* a, const void* b) {
    return (*(int*)a - *(int*)b);
}
longlonggcdSum(int*nums,intnumsSize){

    int*prefixGcd=(int*)malloc(numsSize*sizeof(int));

    intmx=0;

    //buildprefixGcdarray
    for(inti=0;i<numsSize;i++){

        if (nums[i] > mx){
            mx = nums[i];
        }
      prefixGcd[i]=gcd(nums[i],mx);
    }

    qsort(prefixGcd,numsSize,sizeof(int),cmp);
```

```
 longlongsum=0;

    intleft=0;
    int right = numsSize- 1;
    while (left < right) {

        sum+=gcd(prefixGcd[left],prefixGcd[right]);

        left++;
        right--;
    }

    free(prefixGcd);
    return sum;
}
```

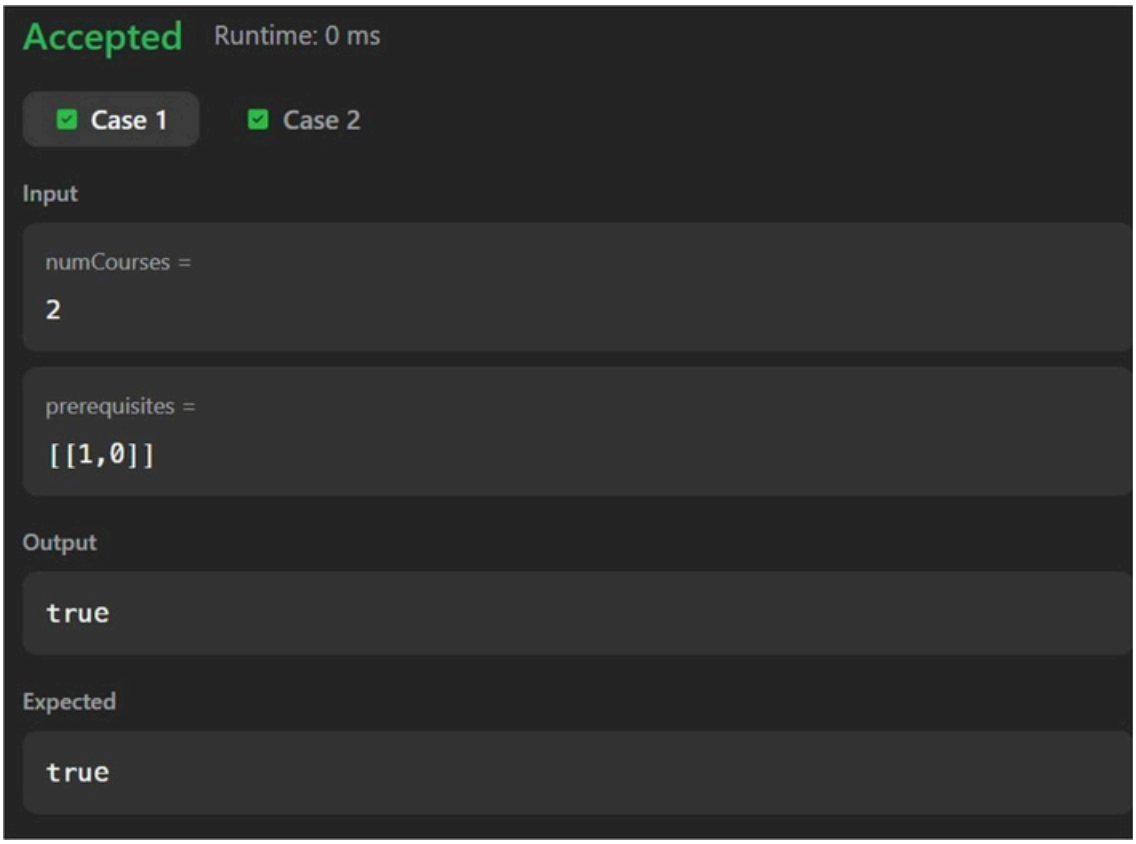


Leetcode 207 : Course Schedule

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {

    int*indegree=(int*)calloc(numCourses,sizeof(int));

    int** graph = (int**)malloc(numCourses * sizeof(int*));
    int* graphSize = (int*)calloc(numCourses, sizeof(int));

    for(inti=0;i<numCourses;i++){
        graph[i]=(int*)malloc(prerequisitesSize*sizeof(int));
    }

    //buildgraph
    for (int i = 0; i < prerequisitesSize; i++) {
        int a = prerequisites[i][0];
        intb=prerequisites[i][1];

        graph[b][graphSize[b]++] = a;
        indegree[a]++;
    }

    int* queue = (int*)malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;

    //pushnodeswithindegree0
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0) {
            queue[rear++]=i;
```

```
    }
}

intcount=0;

while(front<rear){

    int node = queue[front++];
    count++;

    for(inti=0;i<graphSize[node];i++){

        int next = graph[node][i];
        indegree[next]--;

        if (indegree[next] == 0) {
            queue[rear++]=next;
        }
    }
}

for (int i = 0; i < numCourses; i++) {
    free(graph[i]);
}

free(graph);
free(graphSize);
free(indegree);
free(queue);
```

```
    returncount==numCourses;
}
```

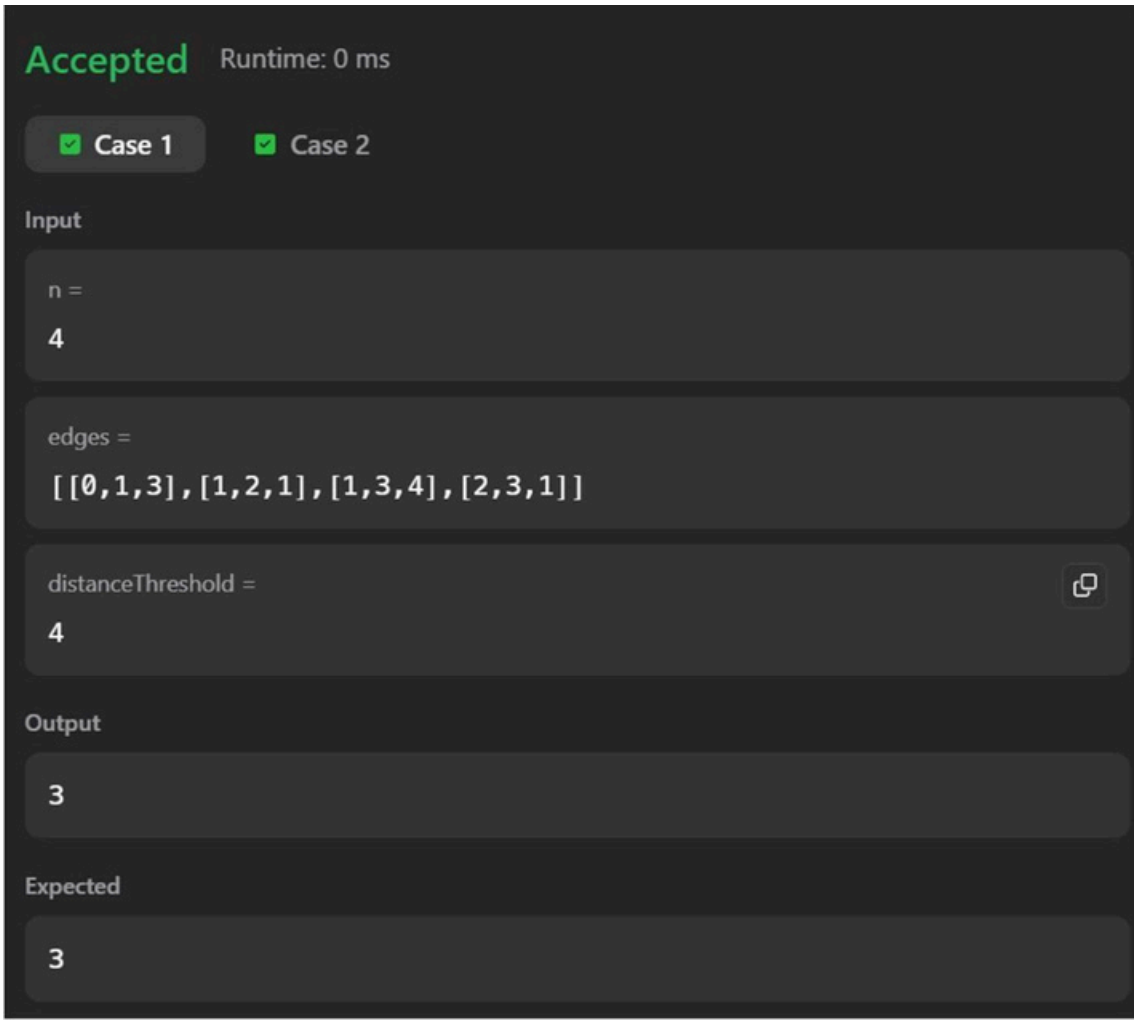


Leetcode 1334 : Find the city with the smallest no of neighbors at a threshold distance

```
#defineINF100000000

int findTheCity(int n, int** edges, int edgesSize, int* edgesColSize, int
distanceThreshold) {

   intdist[101][101];

   for (int i = 0; i < n; i++) {for
       (int j = 0; j < n; j++) {

         if (i == j)
            dist[i][j]=0;
         else
            dist[i][j]=INF;
      }
   }

   for(inti=0;i<edgesSize;i++){

      int  u  =  edges[i][0];
      int  v  =  edges[i][1];
      intw=edges[i][2];

      dist[u][v]=w;
      dist[v][u]=w;
   }

   for(intk=0;k<n;k++){
```

```
    for (int i = 0; i < n; i++) {for
        (int j = 0; j < n; j++) {

                if(dist[i][k]+dist[k][j]<dist[i][j]){
                  dist[i][j]=dist[i][k]+dist[k][j];
            }
        }
    }
}

int answer=-1;
int minReachable=INF;

for(inti=0;i<n;i++){

    intcount=0;

    for(intj=0;j<n;j++){

        if (i != j && dist[i][j] <= distanceThreshold) {
            count++;
        }
    }
    if (count <= minReachable) {
    minReachable = count;
    answer = i;
    }
}
```

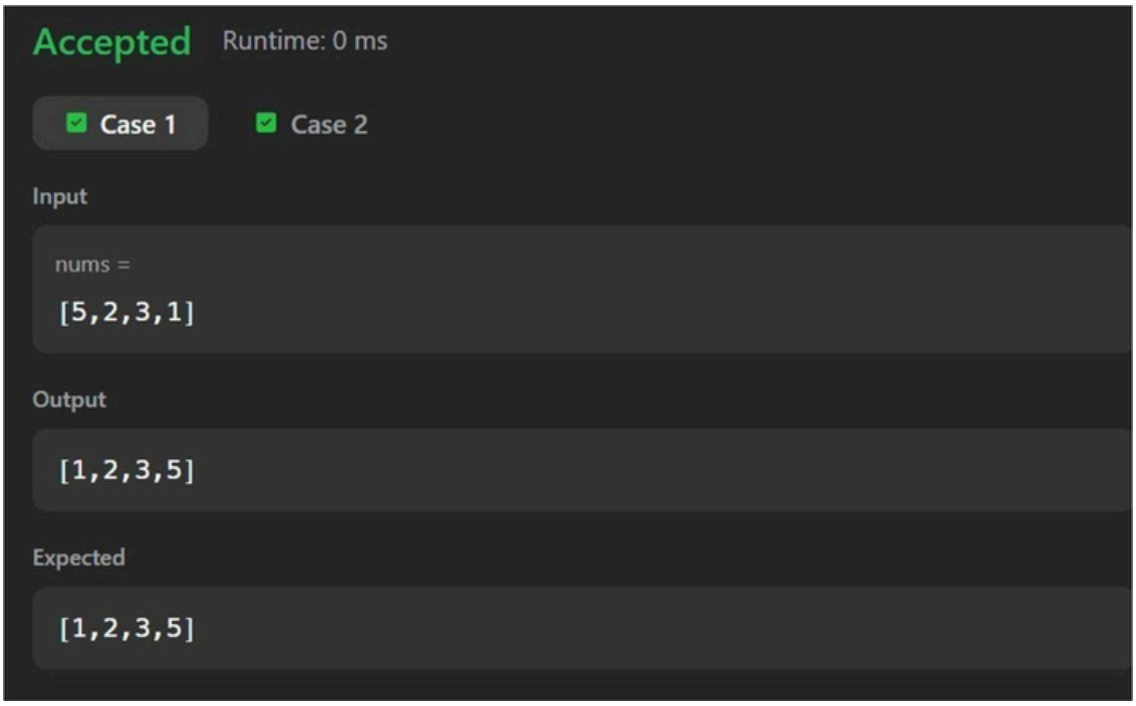
```
    returnanswer;
}
```



Leetcode 912 : Sort an array

```
voidmerge(int*nums,intleft,intmid,intright){

   int n1 = mid - left + 1;
   int n2 = right - mid;

   intL[n1],R[n2];

   for (inti = 0; i < n1; i++)
       L[i] = nums[left + i];

   for (int i = 0; i < n2; i++)
      R[i] = nums[mid + 1 + i];

   inti=0,j=0,k=left;

   while(i<n1&&j<n2){

      if(L[i]<=R[j]){
         nums[k++]=L[i++];
      }else{
         nums[k++]=R[j++];
      }
   }

   while (i < n1) {
      nums[k++] = L[i++];
   }
```

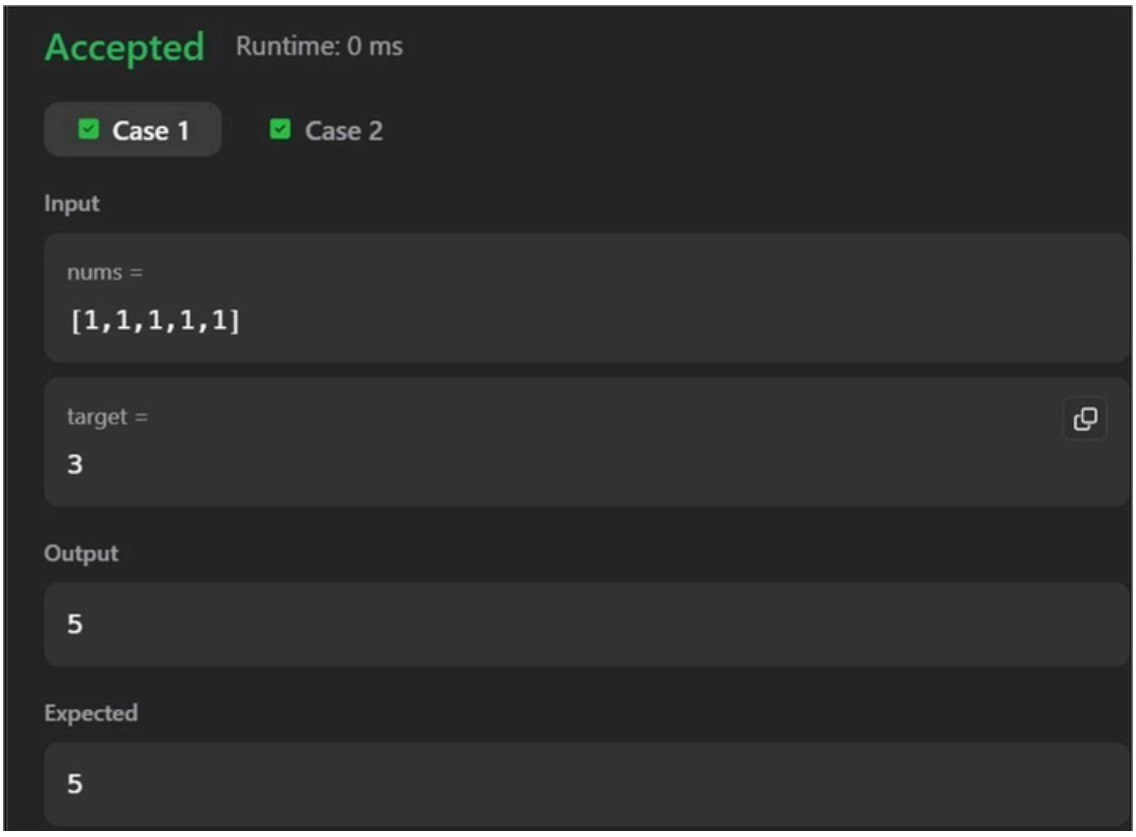
```
    while (j < n2) {
        nums[k++] = R[j++];
    }
}

voidmergeSort(int*nums,intleft,intright){

    if (left>= right)
        return;

    intmid=left+(right-left)/2;

    mergeSort(nums, left, mid);
    mergeSort(nums, mid + 1, right);

    merge(nums,left,mid,right);
}

int*sortArray(int*nums,intnumsSize,int*returnSize){

    mergeSort(nums,0,numsSize-1);

    *returnSize=numsSize;

    returnnums;
}
```


Accepted
Runtime: 0 ms
Case 1
Case 2
Input
nums =
[5,2,3,1]
Output
[1,2,3,5]
Expected
[1,2,3,5]

Leetcode 494 : Target Sum

```
intfindTargetSumWays(int*nums,intnumsSize,inttarget){

    inttotalSum=0;

    for (int i = 0; i < numsSize; i++) {
        totalSum += nums[i];
    }

    //impossiblecase
    if ((target + totalSum) % 2 != 0 || abs(target) > totalSum) {
        return 0;
    }

    intsubsetSum=(target+totalSum)/2;

    intdp[subsetSum+1];

    for (int i = 0; i <= subsetSum; i++) {
        dp[i] = 0;
    }

    dp[0]=1;

    for(inti=0;i<numsSize;i++){

        for (int j = subsetSum; j >= nums[i]; j--) {
            dp[j] += dp[j - nums[i]];
```

```
        }
    }

    returndp[subsetSum];
}
```



Leetcode 316 : Remove duplicate letters

```
char*removeDuplicateLetters(char*s){

   int freq[26] = {0};
   int visited[26] = {0};

   intn=strlen(s);

   for (inti = 0; i < n; i++) {
      freq[s[i] - 'a']++;
   }

   char* stack = (char*)malloc((n + 1) * sizeof(char));
   int top = -1;

   for(inti=0;i<n;i++){

      charch=s[i];

      freq[ch-'a']--;

      if (visited[ch -'a']) {
         continue;
      }

      while (top >= 0 &&
            stack[top] > ch &&
            freq[stack[top]-'a']>0){
```

```
            visited[stack[top] - 'a'] = 0;
            top--;
        }

        stack[++top] = ch;
        visited[ch-'a']=1;
    }

    stack[top+1]='\0';

    returnstack;
}
```